Dynamic rating systems for professional golfers

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Aim of this Project

The aim of my project is to create a rating system for professional golfers, using a data-set provided by Smartodds [1] of all singles golf matches from 2018-2022.



This rating system should...

- produce an individual rating for each competitor, which represents that competitor's skill.
- update the rating of each competitor after they compete in a game.
- be able to predict how likely a competitor is to win their next game depending on their rating and their competitor's ratings.

Traditional Elo-rating update equation

Assume two competitors, i and j, compete against each other and competitor i beats competitor j. Then we would update competitor i's rating, β_i, using **Equation** (1).

$$\beta_i \leftarrow \beta_i + \gamma(X(i\,beats\,j) - \mathbb{E}[X(i\,beats\,j)]), \tag{1}$$

where the learning rate, γ, is a positive constant, $X(i\,beats\,j)$ is 1 if i did win and 0 otherwise. Competitor j's rating, β_j, is updated in the same form as well.

Note that, $(X(i\,beats\,j) - \mathbb{E}[X(i\,beats\,j)])$ is the derivative of the $\log(\mathbb{E}[X(i\,beats\,j))$ with respect to β_i.

The Dataset

The following information about the dataset, Smartodds [1] provided, is relevant to how and when we will update competitor ratings.

- The data-set is data of professional competitions for male golfers from 2018-2022.
- Each competition is split into 4 rounds, where each round is a golf match between many golf competitors.
- For some competitions, after the first 2 rounds, they eliminate the bottom half of competitors from that competition.
- Therefore, it makes the most sense to update competitor's ratings, β, after every round instead of every competition.
- There are 1647 golfers and 398 competitions, so 1592 rounds, in this data-set.

notation

We have already seen that β_i is the competitor's rating, now suppose we have a set, $\mathcal{K}$, of competitors, of length K, then...

- y is the ordered set of competitors in $\mathcal{K}$
- s is the set positions in y

Example: Let $K = 5$, with the competitors labelled $\mathcal{K} = \{1, 2, 3, 4, 5\}$, then if $y = \{1, 2, 4, 5, 3\}$ and $s = \{1, 2, 3, 3, 4\}$, this shows us that competitor 1 came 1st, and competitor 2 came 2nd, but competitors 4 and 5 drew in 3rd place, then competitor 3 came last.

Generalised Plackett-Luce model

We can use a Generalised Plackett-Luce model, GPL [3], to calculate the probability, P, that the events y and s occur. Taking inspiration from Powell's rating system for racers [4], we will split P into stages, where the jth stage represents the probability that a competitor is placed in the jth position, $P^{(j)}$.

Equation (2) shows us the GPL probability equation for P.

$$P = P^{(1)}P^{(2)}...P^{(n-1)} \tag{2}$$

where n is the last number in s, and $P^{(j)}$ is the probability that some competitors win at stage j, where the probability of a stage occurring has the following form in **Equations** 3 and 4.

$$P^{(1)} = \frac{\theta_k \prod_{i \neq \{k\}}(1-\theta_i)}{1-(1-\theta_k)\prod_{i\neq\{k\}}(1-\theta_i)} \tag{3}$$

Equation 3 is the probability competitor k comes 1st, where θ_i is the strength parameter of competitor i.

$$P^{(2)} = \frac{\theta_\ell \prod_{i \neq \{k,\ell\}}(1-\theta_i)}{1-(1-\theta_\ell)\prod_{i\neq\{k,\ell\}}(1-\theta_i)} \tag{4}$$

Equation 4 is the probability competitor ℓ comes 2nd.

You can see that, if a competitor, q, came 3rd, $P^{(3)}$ would be in the same form as $P^{(2)}$ but excluding $\{k, \ell, q\}$ from all $\prod$.

Simple example

If we continue the example from earlier but remove the competitor who came 1st, so $y = \{2, 4, 5, 3\}$ and $s = \{1, 2, 2, 3\}$, then **Equation** (5) is the probability this occurs is.

$$P = P^{(1)}P^{(2)} = \frac{\theta_2(1-\theta_4)(1-\theta_5)(1-\theta_3)}{1-((1-\theta_2)(1-\theta_4)(1-\theta_5)(1-\theta_3))} \frac{\theta_4\theta_5(1-\theta_3)}{1-((1-\theta_4)(1-\theta_5)(1-\theta_3))}, \tag{5}$$

You can see that $P^{(1)}$ includes competitor 2, but, as competitor 2 won the 1st stage, $P^{(2)}$ doesn't include competitor 2 .

Rating, β, update equation

The update equation we will use will have the same form as **Equation** (1), which requires $\frac{\partial \log(P^{(j)})}{\partial \beta_k}$. After finding this, we get **Equation** (6), the update equation for a competitor's rating, β_k.

$$\beta_k^{(1)} \leftarrow \beta_k^{(0)} + \gamma((\mathbb{I}_k^{(j)} - P_k^{(j)}) - \theta_k(1 - P_k^{(j)})) \tag{6}$$

where $\mathbb{I}_k^{(j)} = 1$ if competitor k won at stage j, and 0 if they didn't, and $P_k^{(j)}$ is the probability that competitor k wins in stage j. We set $\beta_k = logit(\theta_k) = log(\theta_k) - log(1-\theta_k)$, therefore $\theta_k = (1 + e^{\beta_k})^{-1}$.

Competitor specific learning rate, γ_k, updates

We will also assign a competitor specific learning rate, γ_k, to help boost competitor's ratings in their first few updates. This will help them reach their predictive rating in fewer updates.

Powell [4] used the negative second derivative of the $\log(P^{(j)})$ to update the reciprocal of γ_k and we will do the same. This gives **Equation** (7).

$$\gamma_k^{-1} \leftarrow \gamma_k^{-1} - \{(1 - P_k^{(j)}) - \theta_k(1 - P_k^{(j)})\}\{(0 - P_k^{(j)}) - \theta_k(1 - P_k^{(j)})\} \tag{7}$$

With this update, every competitor's learning rate, γ_k, will decrease no matter the outcome of the round they compete in. However, the amount their learning rate decreases is dependent on their θ_k and $P_k^{(j)}$.

Results

Using the updates for each competitor's rating, β, and learning rate, γ, in **Equation** (4) and (5), we get the following results, seen in **Figure** 1. Each competitor started with an initial rating of $\beta_0 = -2.03$ and initial learning rate $\gamma_0 = 0.07$. Each competitor's learning rate has the same minimum of $\gamma_{min} = 0.015$, such that $0.015 \leq \gamma_k \leq 0.07$ for all competitors k.

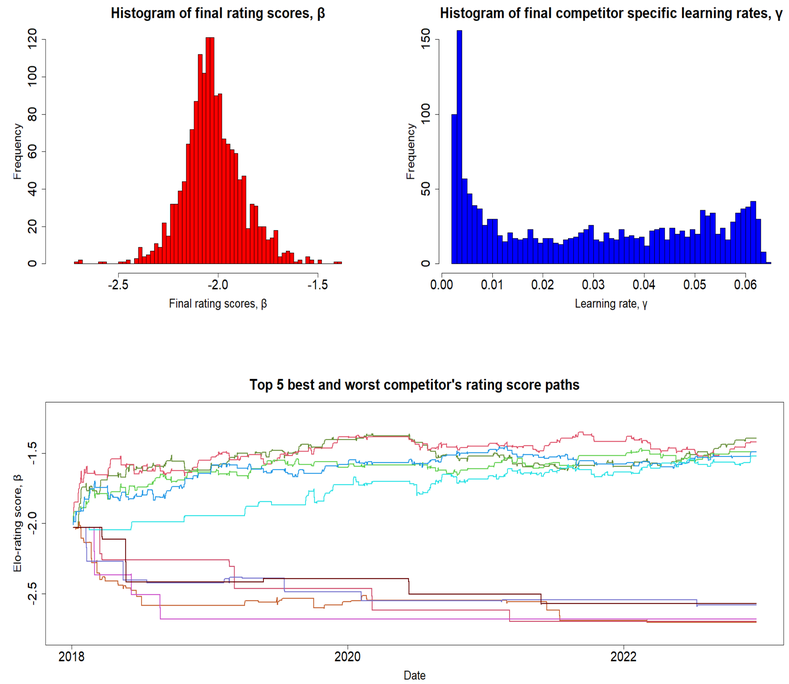


Figure 1

We can compare the top 5 professional golfers, according to our rating system, to the top 5 professional golfers, according to the Official World Golf Rankings (OWGR) [2], in December 2022. **Table** 1 shows this.

Competitor	Rating, β	Ranking according to rating, β	OGWR ranking, December 2022
Rory McIlroy	-1.391	1	1
Jon Rahm	-1.419	2	5
Patrick Cantlay	-1.489	3	4
Xander Schauffele	-1.490	4	6
Scottie Scheffler	-1.521	5	2

Table 1

From the results **Figure** 1 and **Table** 1, we can see that the rating system has fulfilled its three objectives and the results seem broadly consistent with the OWGR [2].

References

[1] https://www.smartodds.co.uk/.

[2] https://www.owgr.com/ranking-graph?minYear=2018&maxYear=2022&player=10091&player=19195 [Accessed: (24/04/2024)].

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